

2.4.1 Continuity Worksheet

1. Which one of the following must hold in order for a function g to be continuous at a number k ?

(a) $\lim_{x \rightarrow \infty} g(x) = g(k)$

(b) $\lim_{x \rightarrow \infty} g(x) = 0$

(c) $\lim_{x \rightarrow \infty} g(x) = k$

(d) $\lim_{x \rightarrow 0} g(x) = g(k)$

(e) $\lim_{x \rightarrow 0} g(x) = 0$

(f) $\lim_{x \rightarrow 0} g(x) = k$

(g) $\lim_{x \rightarrow k} g(x) = g(k)$

(h) $\lim_{x \rightarrow k} g(x) = 0$

(i) $\lim_{x \rightarrow k} g(x) = k$

2. State all values of x where the graph of g has a discontinuity. State the type of discontinuity for each.

$$g(x) = \frac{x^2 - 2x - 8}{2x^2 - 9x + 4}$$

3. Show that the function $f(x) = \begin{cases} \ln(x) + 1 & \text{if } x \leq e \\ (x - e)^2 + 2 & \text{if } x > e \end{cases}$ is continuous at $x = e$.

4. At which points is the following function discontinuous. Justify your answer.

$$f(x) = \begin{cases} (x-1)^2 & \text{if } x \geq 0 \\ x+3 & \text{if } -2 < x < 0 \\ (x+2)^2 + 1 & \text{if } x \leq -2 \end{cases}$$

5. Show that $f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ \sqrt{x} & \text{if } x \geq 0 \end{cases}$ is continuous over $(-\infty, \infty)$.

6. Show that that $f(x) = \begin{cases} \sin(x) & \text{if } x < \pi/4 \\ \cos(x) & \text{if } x \geq \pi/4 \end{cases}$ is continuous over $(-\infty, \infty)$.

7. Find a value for a such that the function $f(x) = \begin{cases} x & \text{if } x \leq a \\ \sqrt{x} & \text{if } x > a \end{cases}$ is continuous at a .

8. Determine the values of c and d so that $f(x) = \begin{cases} 5+x^2 & \text{if } x < -1 \\ cx+d & \text{if } -1 \leq x \leq 5 \\ 7x-11 & \text{if } x > 5 \end{cases}$ is continuous over its domain.